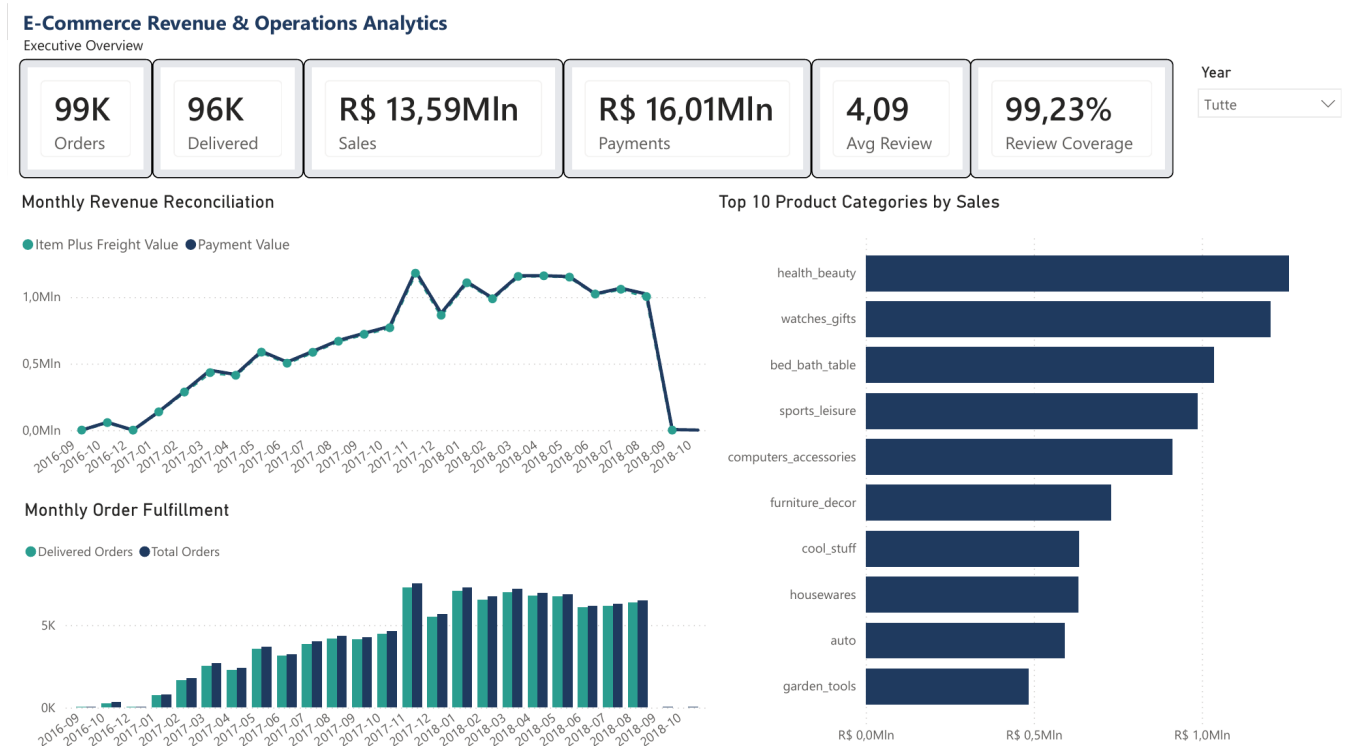


E-Commerce Revenue & Operations Analytics

Portfolio Case Study | Snowflake · dbt · Power BI · Data Quality

Federica Iengo | End-to-end analytics engineering and business intelligence project



28

dbt models

230 / 230

passing regression tests

4

Power BI report pages

3

business marts

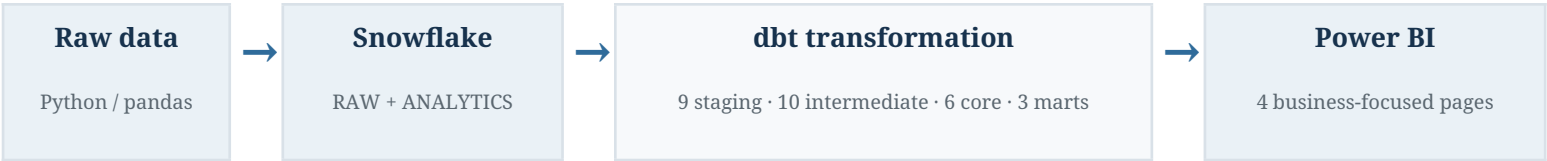
Project summary. A validated analytics layer was built from raw e-commerce transactions and transformed into business-ready dimensional models, reconciliation marts and an interactive Power BI report covering executive performance, financial data quality, delivery operations and customer experience.

Problem, Approach & Architecture

From raw transactional data to a tested analytical layer and business-facing reporting

PROBLEM The source data spans orders, payments, products, sellers, deliveries and reviews, but is not structured for reliable analytical reporting. The core requirement was to build a consistent layer that supports revenue analysis, payment reconciliation, operational monitoring and customer-experience reporting while preserving data-quality issues instead of masking them.	APPROACH • Load and organize source data in Snowflake. • Transform data with dbt across staging, intermediate, core and mart layers. • Model customers, products, sellers, dates, orders and order items for analytical use. • Validate data with automated tests and explicit reconciliation checks. • Connect the validated layer to Power BI for KPI reporting and exploratory analysis.
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Architecture



Validated data model

Dimensions dim_customers · dim_products · dim_sellers · dim_date	Facts fct_orders · fct_order_items	Business marts revenue reconciliation · delivery operations · customer experience
Core row validation 99,441 orders · 112,650 order items	Date integrity 1,338 continuous dates · zero gaps	Regression baseline 230 PASS · 0 WARN · 0 ERROR · 0 SKIP

Business Findings

The project surfaces measurable performance and data-quality signals instead of hiding exceptions

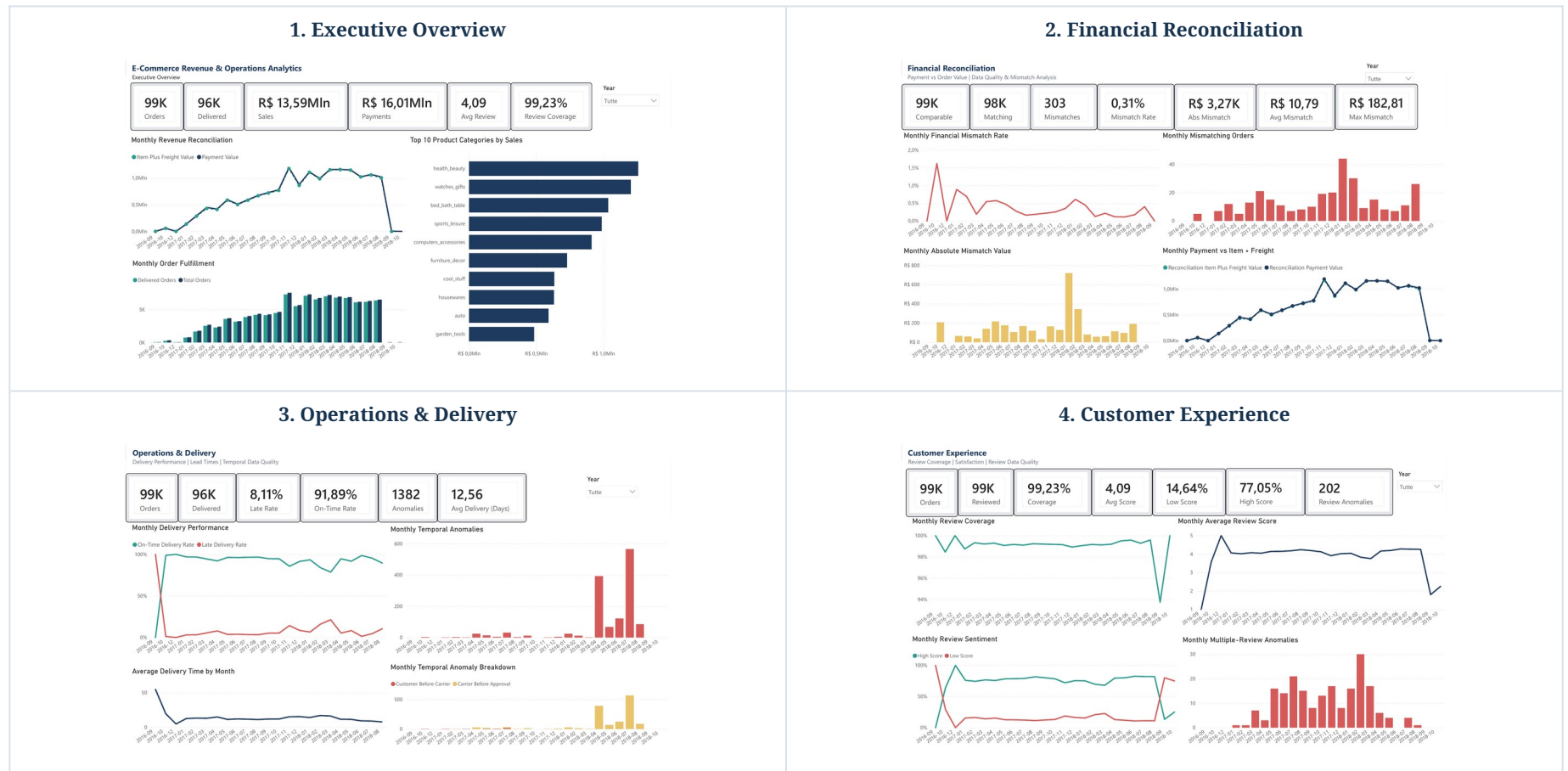
Financial Reconciliation	Operations & Delivery	Customer Experience
98,665 financially comparable orders	96K delivered orders (report KPI)	98,673 reviewed orders
98,362 matching orders	91.89% on-time delivery rate	99.23% review coverage
303 payment / order-value mismatches	8.11% late delivery rate	4.09 / 5 average review score
0.31% mismatch rate	1,382 temporal delivery anomalies	77.05% high-score share
R\$ 3.27K absolute mismatch value	12.56 average delivery days	202 multiple-review anomalies

Why these findings matter

Data quality as a business signal	Reusable analytics layer
Mismatch and temporal anomaly logic is surfaced explicitly, allowing reporting users to distinguish operational performance from source-system quality issues.	The same tested warehouse models support executive, finance, operations and customer-experience views without rebuilding transformation logic inside the BI layer.

Power BI Delivery

Four business-focused views built on the validated Snowflake/dbt analytical layer



Skills demonstrated

Analytics Engineering	Data Modeling	Data Quality	Business Intelligence
dbt layering · reusable marts · warehouse transformations	dimensions · facts · grains · date integrity	reconciliation · anomaly detection · automated regression tests	Power BI · DAX KPIs · filters · stakeholder-oriented reporting

Outcome: a portfolio-ready, end-to-end analytics solution that connects warehouse engineering, validation and business reporting in one defensible workflow.